



PRELIMINARY EXAMINATIONS

HIGHER 2

CANDIDATE
NAME

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CHEMISTRY
9729/02

Paper 2 Structured Questions

23 August 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Civics Group, centre number, index number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 21
2	/ 13
3	/ 9
4	/ 8
5	/ 14
6	/ 10
Total	/ 75

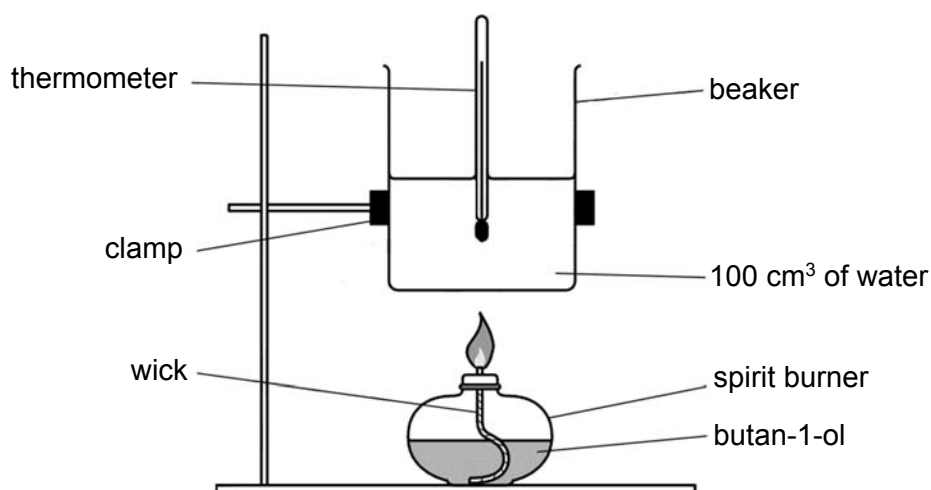
This document consists of **23** printed pages

SECTION A

Answer all questions in the spaces provided.

- 1 (a) For many compounds, the enthalpy change of formation cannot be calculated directly. An indirect method based on enthalpy change of combustion can be used.

The enthalpy change of combustion of butan-1-ol can be found by a calorimetry experiment in which the heat given off during combustion is used to heat a known mass of water and the temperature change recorded.



temperature of water before heating = 25.0 °C

temperature of water after heating = 66.1 °C

mass of spirit burner and butan-1-ol before heating = 80.44 g

mass of spirit burner and butan-1-ol after heating = 79.70 g

- (i) Explain the meaning of the term *standard enthalpy change of combustion*.

.....

.....

..... [1]

- (ii) The heat transfer from the spirit burner to water is known to be only 70% efficient. From the data provided above, determine the enthalpy change of combustion of butan-1-ol.

[2]

- (iii) The entropy change of combustion of butan-1-ol is $-252 \text{ J K}^{-1} \text{ mol}^{-1}$. Using your answer in (a)(ii), calculate the Gibb's free energy for combustion of butan-1-ol at 298 K.

[1]

- (b) The table below gives some data on four fuel sources: methanol, ethanol, hydrogen and octane. Octane can serve as a rough approximation of petrol.

name	formula	molar mass/ g mol ⁻¹	density/ g cm ⁻³	$\Delta H_c^\ominus$ (298K)/ kJ mol ⁻¹	$\Delta H_f^\ominus$ (298K)/ kJ mol ⁻¹
methanol	CH ₃ OH	32	0.793 ^a	-726.0	-239.1
ethanol	CH ₃ CH ₂ OH	46	0.789 ^a	-1367.3	
liquid hydrogen	H ₂	2	0.0711 ^b	-285.8 [#]	
octane	CH ₃ (CH ₂) ₆ CH ₃	114	0.703 ^a	-5470.2	-250.0

^a At 298K and 1 bar pressure

^b At 20K and 1 bar pressure

[#] standard enthalpy change of combustion of hydrogen **gas**

- (i) State the value of the standard enthalpy change of formation of hydrogen **gas**, H₂.

..... [1]

- (ii) Calculate the density of gaseous hydrogen at 298 K and 1 bar pressure. Give your answers in g cm⁻³.

[1]

- (iii) Write the chemical equation that represents the standard enthalpy change of combustion of ethanol, including the state symbols.

.....[1]

- (iv) The standard enthalpy change of combustion of carbon is $-393.5 \text{ kJ mol}^{-1}$. Using this value and the standard enthalpy change of combustion data in the table, calculate the standard enthalpy change of formation of ethanol.

Show your working clearly in the form of an energy cycle diagram.

[3]

- (v) An important property of a fuel, especially when the fuel has to be lifted such as in aviation, is the energy released on combustion per gram of fuel.

Calculate the enthalpy change of combustion per gram of fuel at 1 bar pressure and 298K for methanol.

[1]

- (vi) Another important characteristic of a fuel, especially when there is a fuel tank of limited size, is the energy released on combustion per cm^3 of fuel.

Calculate the enthalpy change of combustion per cm^3 of fuel for octane.

[1]

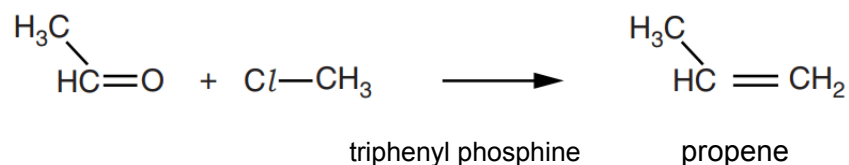
- (vii) Explain why, given the data in the question, it is not strictly possible to make a fair comparison of the energy released per cm^3 of liquid hydrogen with other fuels.

.....

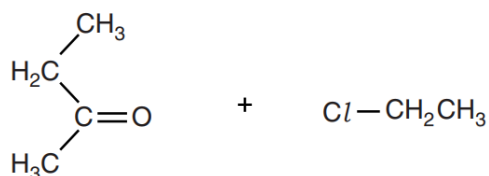
[1]

- (c) Apart from alkanes and alcohols, alkenes are commonly used as fuel. Alkenes can be synthesised through the Wittig reaction, using halogenoalkane and carbonyl compound as the reactants in the presence of triphenyl phosphine.

The equation below shows the synthesis of propene using the Wittig reaction.

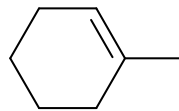


- (i) Draw the structures of the two cis-trans isomers formed from the reaction of the following compounds.



[2]

- (ii) Draw the structural formula of the organic reactant used to generate 1-methylcyclohexene through the Wittig reaction.



1-methylcyclohexene

[1]

Alkenes react readily with interhalogen compound ICl to give halogenalkane. ICl reacts faster with alkenes than pure halogens and can be used to determine the number of carbon-carbon double bonds present in organic compounds.

- (iii) Suggest why ICl reacts with alkenes faster than the pure halogens, Cl_2 , Br_2 and I_2 .

.....
.....[1]

- (iv) Name and describe the mechanism of reaction between propene and ICl to give the major product. In your answer you should show all charges and lone pairs and show the movement of electrons by curly arrows.

[3]

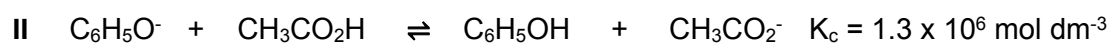
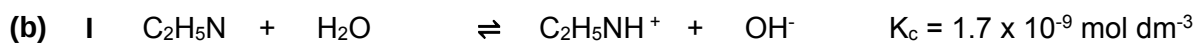
- (v) Draw the pair of enantiomers of the major product from the reaction between propene and ICl .

[1]

[Total: 21]

- 2 (a) What do you understand by the *Brønsted–Lowry* theory of acids and bases?

.....
 [1]



For each of the above equilibrium I and II,

- (i) identify the two acids and the two bases present;

Equilibrium I. Two acids are

Two bases are

Equilibrium II. Two acids are

Two bases are [2]

- (ii) using the given information, suggest, with reasons, which ion or molecule is the stronger acid, and which is the stronger base.

Equilibrium I: stronger acid:

stronger base :

Explanation for Equilibrium I:

.....

Equilibrium II: stronger acid:

stronger base :

Explanation for Equilibrium II:

.....

.....

..... [3]

(c) A student adds an excess of aqueous ethanoic acid to solid calcium carbonate.

(i) Write a full equation for the reaction between ethanoic acid and solid calcium carbonate.

..... [1]

(ii) Explain why a buffer solution has been formed.

.....

..... [1]

(iii) With the aid of an equation, explain how this buffer solution controls pH when an alkali is added.

.....

.....

..... [1]

- (d) A biochemist plans to make up a buffer solution that has a pH of 5.00.

The biochemist adds solid calcium ethanoate to 400 cm³ of 0.200 mol dm⁻³ ethanoic acid. He assumes that the volume of the solution remains constant at 400 cm³ on dissolving the calcium ethanoate.

K_a for ethanoic acid = 1.75×10^{-5} mol dm⁻³

M_r of calcium ethanoate = 158.1

- (i) Calculate the mass of calcium ethanoate that the biochemist needs to dissolve in the ethanoic acid to prepare this buffer solution.

[3]

- (ii) When the biochemist prepares the buffer solution, the volume of solution increases slightly. Suggest whether the pH of the buffer solution would be the same, greater than, or less than pH = 5. Explain your reasoning.

.....

.....

..... [1]

[Total: 13]

- 3 (a) **J**, **K**, **L** and **M** are consecutive elements in the same period of the Periodic Table. The oxide of **J** is basic, oxide of **K** is amphoteric, and the oxides of **L** and **M** are acidic. The halides of **M** can be used to convert alcohols to halogenoalkanes in the absence of water.

(i) Identify the elements **J**, **K**, **L** and **M**.

J: **K:** **L:** **M:**

[1]

(ii) State and explain the variation of atomic radius from **J** to **M**.

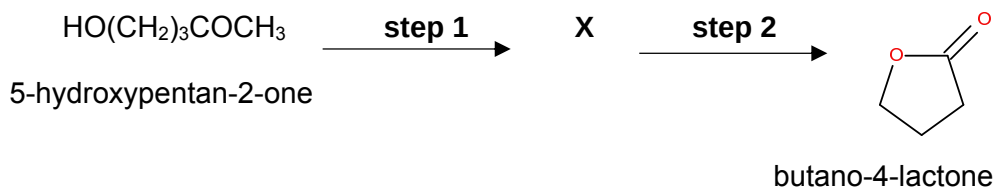
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..... [1]

(iii) Compare the melting points of **L** and **M** and explain your answer.

.....
.....
.....
.....
..... [2]

- (b) Lactones are cyclic esters and are constituents in many natural products used in the flavours and fragrances industry. They also exhibit antioxidant, antimicrobial and anticancer activity.

Butano-4-lactone can be synthesised from 5-hydroxypentan-2-one via a 2-step route as shown.



- (i) State the reagents and conditions for **steps 1** and **2** and identify the intermediate **X** involved.

Step 1 :

.....

Step 2 :

.....

X is

[3]

- (ii) Compound **Y** is a non-cyclic constitutional isomer of butano-4-lactone which exhibits enantiomerism. **Y** gives a reddish brown ppt in an alkaline solution of complexed $\text{Cu}^{2+}(\text{aq})$ and can also liberate carbon dioxide gas upon reaction with acidified KMnO_4 .

Based on the above reactions, identify the functional groups in **Y** and suggest the structural formula of **Y**.

Functional groups in **Y** :

Structural formula of **Y** :

[2]

[Total: 9]

- 4 (a) Air comprises mainly nitrogen and oxygen, with trace quantities of other gases such as argon, carbon dioxide, and even ammonia.

(i) State two assumptions of the kinetic theory of gases.

.....

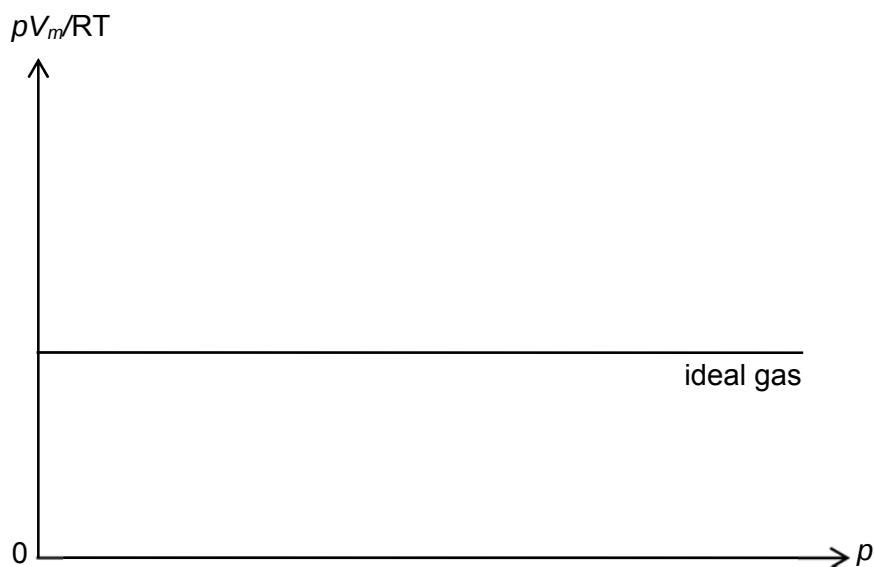
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..... [2]

(ii) A sketch of pV_m/RT against p for 1 mole of an ideal gas at 293 K is given below.



On the same axes, show how 1 mole each of carbon dioxide and ammonia will behave at 293 K. Label your graphs clearly.

[1]

(iii) Briefly explain your answer to **(a)(ii)**.

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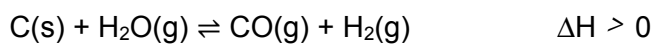
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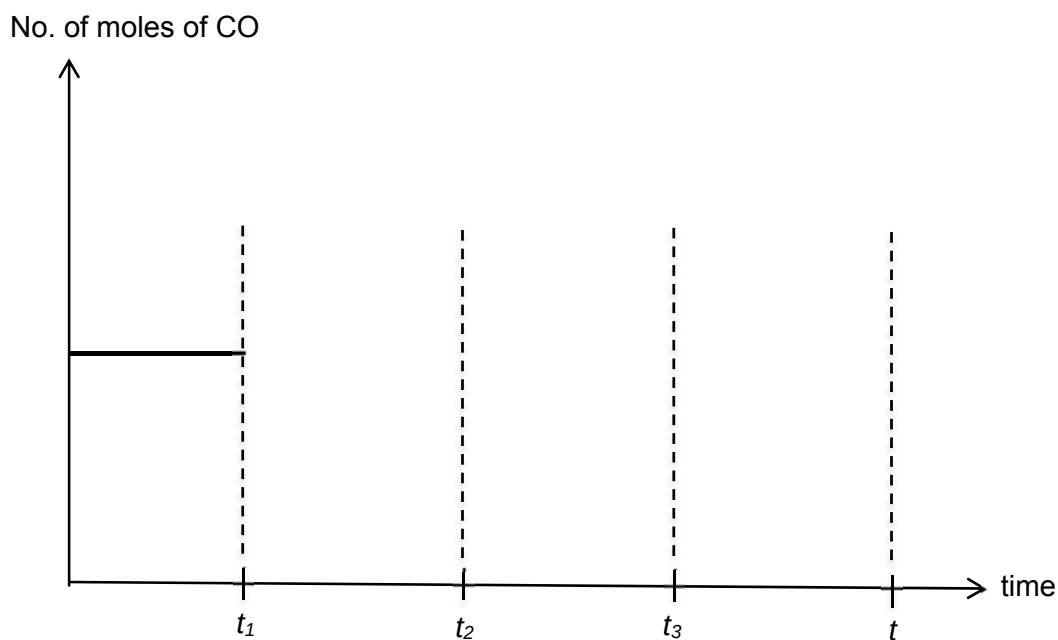
..... [2]

- (b) Water gas is a synthesis gas that comprises carbon monoxide and hydrogen. The gas is made by passing steam over coke:



The system was initially at equilibrium. At time t_1 , the volume of the reaction vessel was suddenly reduced and the system allowed to reach equilibrium. Pressure was then increased by adding argon at constant volume at t_2 , followed by an increase in temperature at t_3 .

Sketch on the axes the graph that should be observed from time = t_1 to time = t when equilibrium is re-established once again.



[3]

[Total:8]

- 5 Chromium is a steely-grey lustrous metal. The name of this element is derived from the Greek word "chrōma" meaning colour, because many of its compounds are intensely coloured.

(a) State and explain **one** difference in physical property between chromium and calcium.

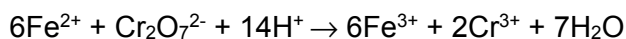
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 [1]

(b) Chromium reacts with oxygen to form a series of oxides.

One of these oxides, chromium (IV) oxide, CrO_2 , is often used to coat data tapes, due to its high conductivity and ferromagnetic properties, which provide a good high audio-frequency response.

When the oxide from a length of tape was dissolved in dilute sulfuric acid, it disproportionates to give $\text{Cr}^{3+}(\text{aq})$ and $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$. The resulting solution needed 20.0 cm^3 of $0.015 \text{ mol dm}^{-3} \text{ Fe}^{2+}$ solution to reduce the $\text{Cr}_2\text{O}_7^{2-}$ completely to Cr^{3+} .



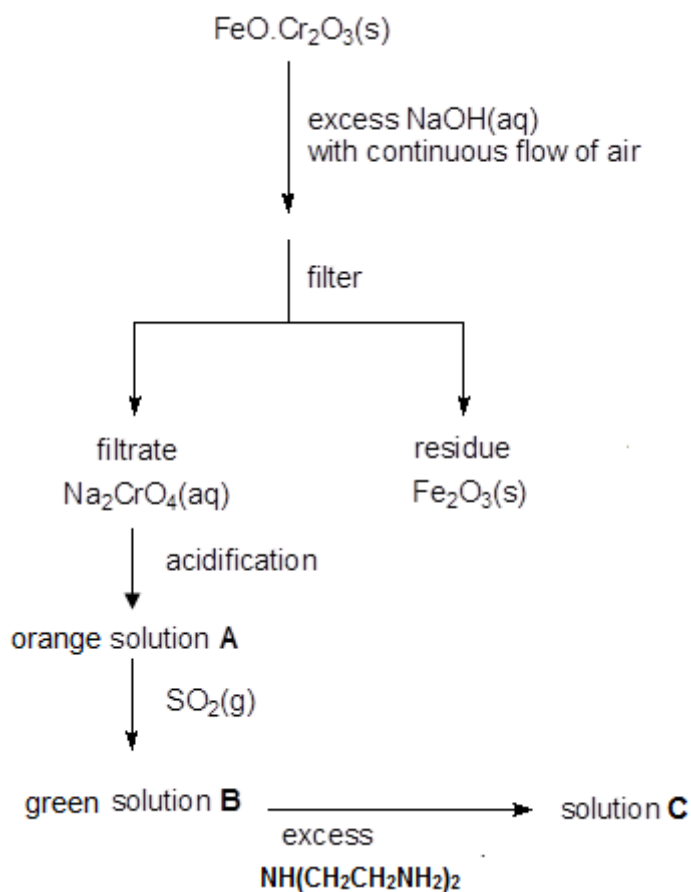
(i) Write an equation for the disproportionation of CrO_2 in acid solution showing how you arrived at the overall equation.

[2]

- (ii) Use the data to calculate the mass of CrO_2 in the length of data tape.

[2]

- (c) Chromite, $\text{FeO} \cdot \text{Cr}_2\text{O}_3$, is the chief source of chromium. The reaction scheme below shows the conversion of chromite.



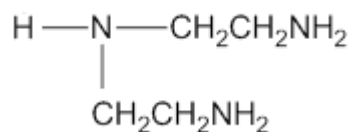
- (i) Identify the chromium containing species present in solution A and in solution B.

Solution A:

Solution B :

[2]

B reacts with excess tri-dentate ligand, diethylenetriamine, $\text{NH}(\text{CH}_2\text{CH}_2\text{NH}_2)_2$, to form **C**.



diethylenetriamine

(ii) What type of reaction occurs when **B** is converted to **C**?

..... [1]

(iii) Letting the tri-dentate ligand be **L**, deduce the formula of the complex ion in **C**.

..... [1]

(iv) Explain why **B** is coloured.

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..... [3]

- (d)** Ag_2CrO_4 is a brown-red crystalline solid and is a chemical precursor to modern photography.

The solubility product of Ag_2CrO_4 is 1.1×10^{-12} .

- (i)** Write an expression for the solubility product, K_{sp} of Ag_2CrO_4 .

..... [1]

- (ii)** Calculate the solubility of Ag_2CrO_4 , in mol dm^{-3} .

[1]

[Total: 14]

6 This question is about the chemistry of Group 17 elements and their compounds.

(a) Although halogens and their compounds can be toxic, some are essential for the human body's functioning and are used in everyday products such as disinfectants and bleaching agents.

(i) State and explain how the oxidising power of the halogens vary down the group.

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.....

.....

.....

.....[2]

(ii) The halogens can react with each other to form interhalogen compounds such as IBr.

IBr reacts with water in which water is acting as the nucleophile. The equation for the reaction is as follows.



State the type of reaction taking place and identify X and Y.

Type of reaction:

X: Y:

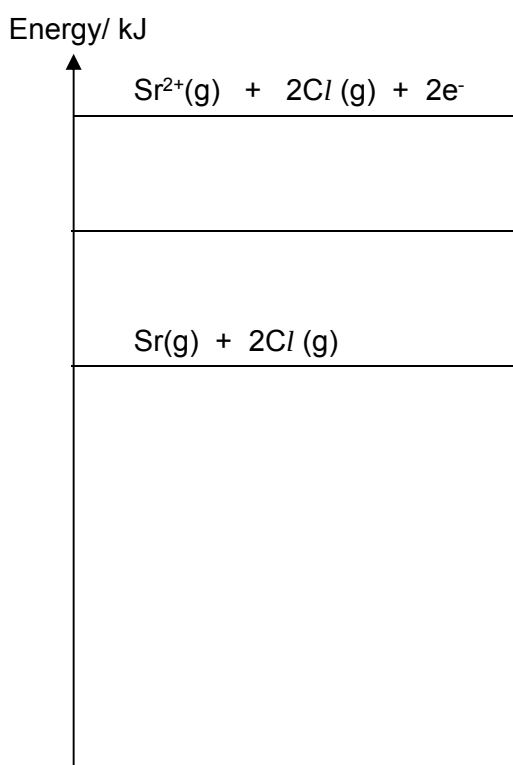
[2]

- (b) (i) Strontium chloride, SrCl_2 is a typical salt which forms neutral aqueous solutions. It emits a bright red colour in a flame which allows it to be used as a source of redness in fireworks.

On the incomplete outline below, construct the energy level diagram to calculate the lattice energy of $\text{SrCl}_2(\text{s})$. Label all enthalpy changes involved and the chemical species at each stage.

Your diagram should include relevant data from the *Data Booklet* together with the following data.

Standard enthalpy change of formation of $\text{SrCl}_2(\text{s})$	$-828.9 \text{ kJ mol}^{-1}$
Standard enthalpy change of atomisation of $\text{Sr}(\text{s})$	$+164.4 \text{ kJ mol}^{-1}$
1 st electron affinity of $\text{Cl}(\text{g})$	$-348.8 \text{ kJ mol}^{-1}$



[3]

- (ii) Using your answer in **(b)(i)**, calculate the $\Delta H^\circ_{\text{soln}}$ of $\text{SrCl}_2(\text{s})$, given the following data:

Standard enthalpy change of hydration of $\text{Sr}^{2+}(\text{g})$	$-1446 \text{ kJ mol}^{-1}$
Standard enthalpy change of hydration of $\text{Cl}^{-}(\text{g})$	-378 kJ mol^{-1}

[1]

- (iii) The standard enthalpy change of solution of $\text{CaCl}_2(\text{s})$ is $-87.7 \text{ kJ mol}^{-1}$.

Explain the difference in the standard enthalpy change of solution of $\text{SrCl}_2(\text{s})$ which you calculated in **(b)(ii)** and that of $\text{CaCl}_2(\text{s})$.

[2]

[Total: 10]